

Viva Dental Prep — NBDHE Board Exam Question Bank

Complete Audited Question Bank with Case Scenarios, Correct Answer Keys & Clinical Rationales

Chapter 9: Microbiology & Immunology | Chapter 10: Infection Control

CHAPTER 9: Microbiology & Immunology

Total Questions: 50 (40 Individual Review Questions + 10 Clinical Case-Based Questions)

Question 1 [CASE-BASED QUESTION]

Case A: Female, 44 years old Chief Complaint "My gums and teeth hurt." Patient History: High blood pressure HIV+, Recovering intravenous drug user Medications: Atenolol (tenormin), Zidovudine (AZT) Last dental visit: five years ago for tooth extraction Current Findings BP: 128/78, Bleeding Index: 55% Generalized necrotic gingival tissues Generalized bone loss Class 2 mobility on teeth 8, 9, 23-26 Generalized bleeding on probing Dental decay #2-O Class V decay #22, 11 Raised, white, nonremovable plaque on dorsum of tongue

The patient's AAP classification based on the information provided would be classified as:

- A. Healthy
- B. Gingivitis
- C. Necrotizing gingivitis
- D. Necrotizing periodontitis ✓ [CORRECT ANSWER]

Correct Answer: Option D

Clinical Rationale: D. The presentation of necrotizing periodontitis includes generalized necrotic tissue with bone loss. _ Healthy gingival tissue appears pink, stippled, firm, with no loss of attachment. _ The presentation of gingivitis includes inflamed gingiva without bone loss. _ The presentation of necrotizing gingivitis includes generalized necrotic tissue without bone loss.

Question 2 [INDIVIDUAL QUESTION]

Which eukaryotic organelle is responsible for protein synthesis?

- A. Centrioles
- B. Cilium
- C. Endoplasmic reticulum ✓ [CORRECT ANSWER]
- D. Lysosome

Correct Answer: Option C

Clinical Rationale: The endoplasmic reticulum (specifically rough ER containing ribosomes) is the eukaryotic organelle responsible for protein synthesis.

Question 3 [CASE-BASED QUESTION]

Case B: Male, 32 years old Chief Complaint: "My lips keep cracking and I need to get my teeth cleaned." Patient History: History of shingles, Recovering drug addict Current diagnoses: anxiety, depression, Type I diabetes Medications: Wellbutrin, Metformin, Lorazepam He has never been to the dentist Current Findings BP: 120/80, Bleeding Index: 50% Generalized 5–7 mm probing depths in the posterior Generalized bone loss, Generalized bleeding on probing Dental decay #2-O, #3-DO, #12-MOD, #19-O, #31-O Class V dental decay #8-11, Mulberry molar presentation on 1st molars Angular cheilitis

The patient's reported history of shingles is MOST likely caused by:

- A. Varicella-zoster virus ✓ [CORRECT ANSWER]
- B. Papillomavirus
- C. Epstein-Barr virus
- D. None of the above

Correct Answer: Option A

Clinical Rationale: A. Varicella zoster virus is the causative agent of shingles.

Question 4 [CASE-BASED QUESTION]

Case A: Female, 44 years old Chief Complaint "My gums and teeth hurt." Patient History: High blood pressure HIV+, Recovering intravenous drug user Medications: Atenolol (tenormin), Zidovudine (AZT) Last dental visit: five years ago for tooth extraction Current Findings BP: 128/78, Bleeding Index: 55% Generalized necrotic gingival tissues Generalized bone loss Class 2 mobility on teeth 8, 9, 23-26 Generalized bleeding on probing Dental decay #2-O Class V decay #22, 11 Raised, white, nonremovable plaque on dorsum of tongue

The AAP classification of this patient commonly occurs in:

- A. Healthy individuals
- B. Individuals undergoing immunosuppressive therapies ✓ [CORRECT ANSWER]
- C. Pregnant individuals
- D. None of the above

Correct Answer: Option B

Clinical Rationale: B. Necrotizing periodontitis commonly occurs in individuals undergoing immunosuppressive therapies.

Question 5 [CASE-BASED QUESTION]

Case B: Male, 32 years old Chief Complaint: "My lips keep cracking and I need to get my teeth cleaned." Patient History: History of shingles, Recovering drug addict Current diagnoses: anxiety, depression, Type I diabetes Medications: Wellbutrin, Metformin, Lorazepam He has never been to the dentist Current Findings BP: 120/80, Bleeding Index: 50% Generalized 5–7 mm probing depths in the posterior Generalized bone loss, Generalized bleeding on probing Dental decay #2-O, #3-DO, #12-MOD, #19-O, #31-O Class V dental decay #8-11, Mulberry molar presentation on 1st molars Angular cheilitis

Angular cheilitis is MOST likely caused by:

- A. S. mutans
- B. C. albicans ✓ [CORRECT ANSWER]
- C. S. pyogenes
- D. C. tetani

Correct Answer: Option B

Clinical Rationale: B. C. albicans is the causative agent of angular cheilitis. ----- S. mutans is the causative agent of dental caries. -- S. pyogenes is the causative agent of streptococcal infections. --- D. C. tetani is the causative agent of tetanus.

Question 6 [INDIVIDUAL QUESTION]

All of the following are signs of inflammation EXCEPT one. Which one is the EXCEPTION?

- A. Bleeding ✓ [CORRECT ANSWER]
- B. Heat
- C. Pain
- D. Redness

Correct Answer: Option A

Clinical Rationale: Bleeding is a clinical sign of tissue damage, but the cardinal signs of inflammation are heat (calor), redness (rubor), pain (dolor), swelling (tumor), and loss of function.

Question 7 [INDIVIDUAL QUESTION]

The type of infection that has a slow onset and long duration is called:

- A. Primary
- B. Focal
- C. Acute
- D. Chronic ✓ [CORRECT ANSWER]

Correct Answer: Option D

Clinical Rationale: A chronic infection has a slow onset and long duration.

Question 8 [CASE-BASED QUESTION]

Case A: Female, 44 years old Chief Complaint "My gums and teeth hurt." Patient History: High blood pressure HIV+, Recovering intravenous drug user Medications: Atenolol (tenormin), Zidovudine (AZT) Last dental visit: five years ago for tooth extraction Current Findings BP: 128/78, Bleeding Index: 55% Generalized necrotic gingival tissues Generalized bone loss Class 2 mobility on teeth 8, 9, 23-26 Generalized bleeding on probing Dental decay #2-O Class V decay #22, 11 Raised, white, nonremovable plaque on dorsum of tongue

The raised, white, nonremovable plaque on the dorsum of tongue is likely:

- A. Kaposi's sarcoma
- B. Hairy leukoplakia
- C. Hyperplastic candidiasis ✓ [CORRECT ANSWER]
- D. Herpetic lesion

Correct Answer: Option C

Clinical Rationale: C. Hyperplastic candidiasis appears as a raised, white, nonremovable plaque on the dorsum of the tongue. - ----A. Kaposi's sarcoma appears as small, reddish blue or purple, or hyperpigmented brown papules, plaques, or nodules on the neck or intraorally. ----B. Hairy leukoplakia presents as white lesions or patches on the tongue. ----D. Herpetic lesions present as fluid-filled, raised vesicles on the lips.

Question 9 [CASE-BASED QUESTION]

Case B: Male, 32 years old Chief Complaint: "My lips keep cracking and I need to get my teeth cleaned." Patient History: History of shingles, Recovering drug addict Current diagnoses: anxiety, depression, Type I diabetes Medications: Wellbutrin, Metformin, Lorazepam He has never been to the dentist Current Findings BP: 120/80, Bleeding Index: 50% Generalized 5–7 mm probing depths in the posterior Generalized bone loss, Generalized bleeding on probing Dental decay #2-O, #3-DO, #12-MOD, #19-O, #31-O Class V dental decay #8-11, Mulberry molar presentation on 1st molars Angular cheilitis

The bacteria MOST likely responsible for the class V decay on #8–11 is:

- A. A. actinomycetecomitans
- B. S. mutans
- C. A. naeslundii ✓ [CORRECT ANSWER]
- D. None of the above

Correct Answer: Option C

Clinical Rationale: C. A. naeslundii is primarily responsible for root surface caries. ----- . A. actinomycetecomitans is the causative organism of periodontitis. - . S. mutans is responsible for the initiation of the caries process.

Question 10 [INDIVIDUAL QUESTION]

The human immunodeficiency virus (HIV) can be transmitted through all of the following routes EXCEPT one. Which one is the EXCEPTION?

- A. Sharing utensils with an HIV positive person ✓ [CORRECT ANSWER]
- B. Intravenous drug use
- C. Sexual contact with an HIV positive person
- D. Perinatal transmission

Correct Answer: Option A

Clinical Rationale: HIV is NOT transmitted through casual contact such as sharing eating utensils. It is transmitted via blood, sexual contact, IV drug use, and perinatal exposure.

Question 11 [CASE-BASED QUESTION]

Case A: Female, 44 years old Chief Complaint "My gums and teeth hurt." Patient History: High blood pressure HIV+, Recovering intravenous drug user Medications: Atenolol (tenormin), Zidovudine (AZT) Last dental visit: five years ago for tooth extraction Current Findings BP: 128/78, Bleeding Index: 55% Generalized necrotic gingival tissues Generalized bone loss Class 2 mobility on teeth 8, 9, 23-26 Generalized bleeding on probing Dental decay #2-O Class V decay #22, 11 Raised, white, nonremovable plaque on dorsum of tongue

44. The decay on #2-O is likely caused by which organism?

- A. actinomycetecomitans
- B. S. mutans ✓ [CORRECT ANSWER]
- C. A. naeslundii
- D. None of the above

Correct Answer: Option B

Clinical Rationale: B. S. mutans is the causative agent of dental caries.

Question 12 [CASE-BASED QUESTION]

Case B: Male, 32 years old Chief Complaint: "My lips keep cracking and I need to get my teeth cleaned." Patient History: History of shingles, Recovering drug addict Current diagnoses: anxiety, depression, Type I diabetes Medications: Wellbutrin, Metformin, Lorazepam He has never been to the dentist Current Findings BP: 120/80, Bleeding Index: 50% Generalized 5–7 mm probing depths in the posterior Generalized bone loss, Generalized bleeding on probing Dental decay #2-O, #3-DO, #12-MOD, #19-O, #31-O Class V dental decay #8-11, Mulberry molar presentation on 1st molars Angular cheilitis

The most likely cause of the patient's mulberry molar presentation is:

- A. Syphilis ✓ [CORRECT ANSWER]
- B. HIV
- C. Gonorrhea
- D. Infective endocarditis

Correct Answer: Option A

Clinical Rationale: A. Mulberry molars are an oral manifestation of congenital syphilis. -----Oral manifestations of HIC include hairy leukoplakia, herpetic lesions, recurrent aphthous ulcers, candidiasis, papillomavirus, gingival erythema, necrotizing periodontitis. -----Gonococcal glossitis is a manifestation of gonorrhea.

Question 13 [CASE-BASED QUESTION]

Case B: Male, 32 years old Chief Complaint: "My lips keep cracking and I need to get my teeth cleaned." Patient History: History of shingles, Recovering drug addict Current diagnoses: anxiety, depression, Type I diabetes Medications: Wellbutrin, Metformin, Lorazepam He has never been to the dentist Current Findings BP: 120/80, Bleeding Index: 50% Generalized 5–7 mm probing depths in the posterior Generalized bone loss, Generalized bleeding on probing Dental decay #2-O, #3-DO, #12-MOD, #19-O, #31-O Class V dental decay #8-11, Mulberry molar presentation on 1st molars Angular cheilitis

The medication that is indicated for diabetes is:

- A. Lorazepam
- B. Wellbutrin
- C. Metformin ✓ [CORRECT ANSWER]
- D. None of the above

Correct Answer: Option C

Clinical Rationale: C. Metformin is indicated for diabetes. -----A. Lorazepam is indicated for anxiety.

Question 14 [CASE-BASED QUESTION]

Case A: Female, 44 years old Chief Complaint "My gums and teeth hurt." Patient History: High blood pressure HIV+, Recovering intravenous drug user Medications: Atenolol (tenormin), Zidovudine (AZT) Last dental visit: five years ago for tooth extraction Current Findings BP: 128/78, Bleeding Index: 55% Generalized necrotic gingival tissues Generalized bone loss Class 2 mobility on teeth 8, 9, 23-26 Generalized bleeding on probing Dental decay #2-O Class V decay #22, 11 Raised, white, nonremovable plaque on dorsum of tongue

HIV was most likely transmitted by:

- A. Sharing utensils with an HIV-infected individual
- B. Hugging an HIV-infected individual
- C. Using the same restroom as an HIV-infected individual
- D. Intravenous drug use ✓ [CORRECT ANSWER]

Correct Answer: Option D

Clinical Rationale: D. HIV is transmitted through intravenous drug use.

Question 15 [INDIVIDUAL QUESTION]

All of the following are oral manifestations of human immunodeficiency virus EXCEPT one. Which one is the EXCEPTION?

- A. Recurrent aphthous ulcers
- B. Warts ✓ [CORRECT ANSWER]
- C. Kaposi's sarcoma
- D. Hairy leukoplakia

Correct Answer: Option B

Clinical Rationale: Oral warts (caused by Human Papillomavirus, HPV) are not classified as classic opportunistic HIV lesions like Kaposi sarcoma, hairy leukoplakia, or aphthous ulcers.

Question 16 [INDIVIDUAL QUESTION]

The causative agent of warts is:

- A. Epstein-Barr virus
- B. Neisseria gonorrhoeae
- C. Papillomaviruses ✓ [CORRECT ANSWER]
- D. Rubella virus

Correct Answer: Option C

Clinical Rationale: Human Papillomaviruses (HPV) are the causative agents of oral and cutaneous warts.

Question 17 [INDIVIDUAL QUESTION]

All of the following are prokaryotic EXCEPT one. Which one is the EXCEPTION?

- A. Fungi ✓ [CORRECT ANSWER]
- B. Viruses
- C. Eubacteria
- D. Archeobacteria

Correct Answer: Option A

Clinical Rationale: Fungi are eukaryotic organisms (having a membrane-bound nucleus and organelles). Bacteria and archaea are prokaryotic.

Question 18 [INDIVIDUAL QUESTION]

Which type of observation allows specimens to be seen as bright objects against a dark background?

- A. Phase-contrast microscopy
- B. Bright-field microscopy
- C. Fluorescence microscopy
- D. Darkfield microscopy ✓ [CORRECT ANSWER]

Correct Answer: Option D

Clinical Rationale: Darkfield microscopy illuminates specimens so they appear bright against a dark background, ideal for viewing live spirochetes like *Treponema pallidum*.

Question 19 [INDIVIDUAL QUESTION]

The type of immunity a child may receive from the antibodies of the mother is called:

- A. Cell-mediated immunity
- B. Passive acquired immunity ✓ [CORRECT ANSWER]
- C. Active acquired immunity
- D. Humoral immunity

Correct Answer: Option B

Clinical Rationale: Passive acquired immunity occurs when preformed antibodies are transferred from mother to child via placenta (IgG) or breast milk (IgA).

Question 20 [INDIVIDUAL QUESTION]

The first immune cell to respond to infection are called:

- A. B-lymphocytes
- B. T-lymphocytes
- C. Polymorphonuclear leukocytes ✓ [CORRECT ANSWER]
- D. None of the above

Correct Answer: Option C

Clinical Rationale: Polymorphonuclear leukocytes (PMNs / Neutrophils) are the first immune cells to migrate to the site of infection/inflammation.

Question 21 [INDIVIDUAL QUESTION]

The presence of toxins in the bloodstream is called:

- A. Bacteremia
- B. Toxemia ✓ [CORRECT ANSWER]
- C. Nosocomial infections
- D. None of the above

Correct Answer: Option B

Clinical Rationale: Toxemia is the presence of circulating toxins in the bloodstream.

Question 22 [INDIVIDUAL QUESTION]

Which type of staining procedure determines the presence of prions, proteoglycans, and glycosaminoglycans present in tissue?

- A. Negative staining
- B. Schaeffer-Fulton staining
- C. Flagellar staining
- D. Toluidine blue staining ✓ [CORRECT ANSWER]

Correct Answer: Option D

Clinical Rationale: Toluidine blue staining is used to highlight prions, proteoglycans, and glycosaminoglycans in tissue specimens.

Question 23 [INDIVIDUAL QUESTION]

Bacteria that appear rod-like and occur in pairs is called:

- A. Streptobacilli
- B. Diplobacilli ✓ [CORRECT ANSWER]**
- C. Coccobacilli
- D. Fusiform bacilli

Correct Answer: Option B

Clinical Rationale: Diplobacilli are rod-shaped bacteria occurring in pairs.

Question 24 [INDIVIDUAL QUESTION]

Bacteria that appear spherical or ovoid and occur in irregular clusters are called:

- A. Staphylococci ✓ [CORRECT ANSWER]**
- B. Streptococci
- C. Diplococci
- D. Tetrad

Correct Answer: Option A

Clinical Rationale: Staphylococci are spherical or ovoid bacteria that divide in multiple planes to form irregular grape-like clusters.

Question 25 [INDIVIDUAL QUESTION]

All of the following are types of lower respiratory bacterial infections EXCEPT one. Which one is the EXCEPTION?

- A. Tuberculosis
- B. Bacterial pneumonia
- C. Whooping cough
- D. Coronavirus disease (COVID-19) ✓ [CORRECT ANSWER]**

Correct Answer: Option D

Clinical Rationale: COVID-19 (caused by SARS-CoV-2) is a VIRAL lower respiratory infection. Tuberculosis, bacterial pneumonia, and pertussis are bacterial.

Question 26 [INDIVIDUAL QUESTION]

A common oral manifestation of Scarlet fever is:

- A. Hairy leukoplakia
- B. Herpetic lesions
- C. Strawberry tongue ✓ [CORRECT ANSWER]**
- D. Human papillomavirus

Correct Answer: Option C

Clinical Rationale: A red, hyperemic strawberry tongue (followed by a white coated tongue) is a pathognomonic oral sign of Scarlet Fever caused by *Streptococcus pyogenes*.

Question 27 [INDIVIDUAL QUESTION]

Which bacteria cause syphilis?

- A. *Treponema pallidum* ✓ [CORRECT ANSWER]
- B. *Streptococcus mutans*
- C. *Clostridium tetani*
- D. None of the above

Correct Answer: Option A

Clinical Rationale: *Treponema pallidum* (an anaerobic spirochete) is the causative bacterium for syphilis.

Question 28 [INDIVIDUAL QUESTION]

The type of microbial relationship in which two different species live in a mutually beneficial coexistence is called:

- A. Commensalism
- B. Symbiosis ✓ [CORRECT ANSWER]
- C. Predation
- D. Parasitism

Correct Answer: Option B

Clinical Rationale: Symbiosis is a close microbial relationship in which two species live in a mutually beneficial coexistence.

Question 29 [INDIVIDUAL QUESTION]

Which immunoglobulin (Ig) plays a major role in hypersensitivity reactions?

- A. IgM
- B. IgG
- C. IgA
- D. IgE ✓ [CORRECT ANSWER]

Correct Answer: Option D

Clinical Rationale: IgE is the key immunoglobulin mediating Type I hypersensitivity (allergic) reactions and parasitic defense.

Question 30 [INDIVIDUAL QUESTION]

A gumma is present during which stage of syphilis infection?

- A. Primary
- B. Secondary
- C. Tertiary ✓ [CORRECT ANSWER]
- D. Quaternary

Correct Answer: Option C

Clinical Rationale: A gumma (granulomatous lesion) is characteristic of Tertiary (third stage) syphilis.

Question 31 [INDIVIDUAL QUESTION]

What is the causative agent for warts?

- A. Papillomavirus ✓ [CORRECT ANSWER]
- B. Rubeola virus
- C. Varicella-zoster virus
- D. Orthomyxovirus

Correct Answer: Option A

Clinical Rationale: Human Papillomavirus (HPV) is the causative viral agent of warts.

Question 32 [INDIVIDUAL QUESTION]

The immunity acquired when a person is exposed to an antigen and the body produces antibodies is called:

- A. Natural—passive
- B. Natural—active ✓ [CORRECT ANSWER]
- C. Artificial—active
- D. Artificial—passive

Correct Answer: Option B

Clinical Rationale: Natural active immunity develops when an individual is exposed to a live pathogen, develops the disease, and produces immune memory antibodies.

Question 33 [INDIVIDUAL QUESTION]

Which agent causes infectious mononucleosis?

- A. Papillomavirus
- B. Rubeola virus
- C. Varicella-zoster virus
- D. Epstein-Barr virus ✓ [CORRECT ANSWER]

Correct Answer: Option D

Clinical Rationale: Epstein-Barr virus (EBV / HHV-4) is the causative agent of infectious mononucleosis.

Question 34 [INDIVIDUAL QUESTION]

Enamel demineralization typically occurs at which pH?

- A. 2.0–3.0
- B. 4.5–5.5 ✓ [CORRECT ANSWER]
- C. 6.0–6.7
- D. None of the above

Correct Answer: Option B

Clinical Rationale: Enamel demineralization begins when critical pH drops to 4.5 – 5.5.

Question 35 [INDIVIDUAL QUESTION]

Root-surface demineralization typically occurs at which pH?

- A. 2.0–3.0
- B. 4.5–5.5
- C. 6.0–6.7 ✓ [CORRECT ANSWER]
- D. None of the above

Correct Answer: Option C

Clinical Rationale: Root surface (cementum/dentin) demineralization occurs at a higher critical pH of 6.0 – 6.7.

Question 36 [INDIVIDUAL QUESTION]

All of the following organisms are colonizers of soft tissues EXCEPT one. Which one is the EXCEPTION?

- A. *Streptococcus mutans* ✓ [CORRECT ANSWER]
- B. *Streptococcus mitis*
- C. *Candida albicans*
- D. *Streptococcus salivarius*

Correct Answer: Option A

Clinical Rationale: *Streptococcus mutans* colonizes hard tooth surfaces (enamel), whereas *S. mitis* and *S. salivarius* colonize soft mucosal tissues.

Question 37 [INDIVIDUAL QUESTION]

A 43-year-old male presented for his dental prophylaxis. While conducting the oral cancer examination, the dental hygienist noticed a smooth, flat, red lesion on the dorsum of the tongue. Which of the following fungal infections is likely present?

- A. Pseudomembranous candidiasis
- B. Erythematous candidiasis ✓ [CORRECT ANSWER]
- C. Angular cheilitis
- D. Hyperplastic candidiasis

Correct Answer: Option B

Clinical Rationale: Erythematous candidiasis presents as a smooth, flat, red macule on the dorsum of the tongue.

Question 38 [INDIVIDUAL QUESTION]

The type of organism that oxidizes chemical compounds for energy is called:

- A. Phototrophic
- B. Hypotrophic
- C. Chemotrophic ✓ [CORRECT ANSWER]
- D. None of the above

Correct Answer: Option C

Clinical Rationale: Chemotrophic organisms obtain energy by oxidizing chemical compounds.

Question 39 [INDIVIDUAL QUESTION]

All of the following are periodontal pathogens EXCEPT one. Which one is the EXCEPTION?

- A. *P. gingivalis*
- B. *T. forsythia*
- C. *S. sanguinis* ✓ [CORRECT ANSWER]
- D. *A. actinomycetemcomitans*

Correct Answer: Option C

Clinical Rationale: *Streptococcus sanguinis* is a pioneer supragingival colonizer, not a red/orange complex periodontal pathogen.

Question 40 [INDIVIDUAL QUESTION]

A 75-year-old woman presents for her periodontal maintenance procedure. When asked to open her mouth, the hygienist noticed cracking and redness at the commissures of the patient's lip. Which of the following conditions is likely present?

- A. Angular cheilitis ✓ [CORRECT ANSWER]
- B. Aphthous ulcer
- C. Herpes
- D. None of the above

Correct Answer: Option A

Clinical Rationale: Angular cheilitis presents as painful erythematous cracking and fissuring at the commissures of the lips.

Question 41 [INDIVIDUAL QUESTION]

The predominant microflora most likely to be found in supragingival plaque are:

- A. Gram-positive cocci and rods ✓ [CORRECT ANSWER]
- B. Gram-negative cocci and rods
- C. Spirochetes
- D. None of the above

Correct Answer: Option A

Clinical Rationale: Early supragingival plaque is dominated by Gram-positive cocci and rods (e.g., *Streptococcus* and *Actinomyces* species).

Question 42 [INDIVIDUAL QUESTION]

All of the following describe cytoplasm EXCEPT one. Which one is the EXCEPTION?

- A. Cytoplasm is the fluid compartment inside of the cytoplasmic membrane.
- B. Cytoplasm is the prominent site for many of the cell's biochemical and synthetic pathways.
- C. Cytoplasm contains chromatin body, ribosomes, and granules.
- D. Cytoplasm determines and maintains the shape of the microorganism. ✓ [CORRECT ANSWER]

Correct Answer: Option D

Clinical Rationale: Cell shape is determined and maintained by the bacterial cell wall (peptidoglycan), not the cytoplasm.

Question 43 [INDIVIDUAL QUESTION]

All of the following structures are found in a eukaryotic cell EXCEPT one. Which one is the EXCEPTION?

- A. Nucleoid ✓ [CORRECT ANSWER]
- B. Nucleus
- C. Mitochondria
- D. Golgi complex

Correct Answer: Option A

Clinical Rationale: A nucleoid (non-membrane-bound genetic region) is found in prokaryotic cells. Eukaryotic cells possess a true membrane-bound nucleus.

Question 44 [INDIVIDUAL QUESTION]

When a causative agent remains inactive for a period of time but later becomes active and produces symptoms of disease is called:

- A. Nosocomial
- B. Bacterimia
- C. Latent ✓ [CORRECT ANSWER]
- D. Acute

Correct Answer: Option C

Clinical Rationale: A latent infection remains dormant in host tissues without active replication before reactivating later to produce disease symptoms.

Question 45 [INDIVIDUAL QUESTION]

A herpetic infection that occurs on the finger is commonly called:

- A. Herpes genitalis
- B. Herpetic whitlow ✓ [CORRECT ANSWER]
- C. Herpes labialis
- D. None of the above

Correct Answer: Option B

Clinical Rationale: Herpetic whitlow is a primary or secondary HSV infection occurring on the fingers or hands.

Question 46 [INDIVIDUAL QUESTION]

The causative agent in chickenpox is:

- A. Varicella-zoster virus ✓ [CORRECT ANSWER]
- B. Herpes simplex virus
- C. Rubella virus
- D. Influenza virus

Correct Answer: Option A

Clinical Rationale: Varicella-Zoster Virus (VZV / HHV-3) causes chickenpox (primary infection) and shingles (reactivation).

Question 47 [INDIVIDUAL QUESTION]

What type of hepatitis is a dental professional most at risk for contracting through an occupational bloodborne exposure?

- A. Hepatitis A
- B. Hepatitis B ✓ [CORRECT ANSWER]**
- C. Hepatitis E
- D. None of the above

Correct Answer: Option B

Clinical Rationale: Hepatitis B Virus (HBV) is the bloodborne virus posing the greatest risk of transmission to unvaccinated dental healthcare workers via percutaneous injuries.

Question 48 [INDIVIDUAL QUESTION]

All of the following hepatitis viruses result in chronic liver disease EXCEPT one. Which one is the EXCEPTION?

- A. Hepatitis A ✓ [CORRECT ANSWER]**
- B. Hepatitis B
- C. Hepatitis C
- D. Hepatitis D

Correct Answer: Option A

Clinical Rationale: Hepatitis A is an acute, self-limiting fecal-oral infection that does NOT cause chronic liver disease.

Question 49 [INDIVIDUAL QUESTION]

When performing a dental prophylaxis, a patient with infective endocarditis is most at risk for:

- A. bacteremia ✓ [CORRECT ANSWER]**
- B. nosocomial infection
- C. toxemia
- D. none of the above

Correct Answer: Option A

Clinical Rationale: Bacteremia (transient entry of oral bacteria into the bloodstream) occurs during dental prophylaxis, posing a risk of infective endocarditis in vulnerable patients.

Question 50 [INDIVIDUAL QUESTION]

All of the following are prerequisites for caries development EXCEPT one. Which one is the EXCEPTION?

- A. Streptococcus mutans
- B. Candida albicans ✓ [CORRECT ANSWER]**
- C. Susceptible host/tooth
- D. Sugar substrate

Correct Answer: Option B

Clinical Rationale: Candida albicans is a fungal organism and is NOT a prerequisite for dental caries. Caries requires a susceptible host, cariogenic bacteria, fermentable carbohydrates, and time.

CHAPTER 10: Infection Control

Total Questions: 50 (42 Individual Review Questions + 8 Clinical Case-Based Questions)

Question 1 [CASE-BASED QUESTION]

Case A: Jean, a registered dental hygienist (RDH) working through a temporary agency, is assigned a shift at Bright Smile Dental office. When she arrived, Jean was shown her operatory, schedule, and instruments. She was given a brief tour of the office to show her the radiology room, the sterilization lab, and the break room. No one explained any standard operating procedures (SOPs) or showed her where the OSHA manual was located. As Jean was unloading the autoclave, she noticed the instrument pouches were still wet because another staff member opened the door early because she needed extraction instruments right away.

What should Jean do next?

- A. Go ahead and use the wet instruments.
- B. Wait for the instruments to dry and then use the instruments.
- C. Close the autoclave and restart the cycle.
- D. Rebag the instruments and run a new cycle including a drying cycle. ✓ [CORRECT ANSWER]

Correct Answer: Option D

Clinical Rationale: Re-bag the instruments and run a new cycle including a drying cycle. Must be re-packaged with a new pouch so the indicator changes color and re-sterilized with a full drying cycle.

Question 2 [INDIVIDUAL QUESTION]

Which of these groups is a regulatory body?

- A. The Occupational Safety and Health Administration (OSHA) ✓ [CORRECT ANSWER]
- B. The Centers for Disease Control and Prevention (CDC)
- C. The Association for Dental Safety (ADS)
- D. The American Dental Association (ADA)

Correct Answer: Option A

Clinical Rationale: OSHA is a federal regulatory agency under the US Department of Labor that formulates and enforces workplace safety regulations.

Question 3 [CASE-BASED QUESTION]

Case B: Pedro, a registered dental hygienist (RDH), was recently hired at Oceanside Dental Group. On his first day, Pedro was being trained by the lead dental hygienist and Infection Control Coordinator, Sarah. Sarah made sure to explain the schedule, reviewed the OSHA manual, reviewed the walkout procedures, and reviewed all of the infection control policies and procedures located in the Standard Operating Procedures (SOP) manual. As Pedro was turning his treatment room over after dismissing his first patient, he forgot a few things Sarah said regarding the contact time for the disinfectant spray and where the water line treatment tablets were located. He asked another staff member, Cindy, about these questions. Cindy replied, "Sarah goes overboard with infection control. We don't listen to her. Our schedules are so full that we don't have time to do all that stuff." What should Pedro do?

How long should Pedro wait to seat his next patient after cleaning and disinfecting his operatory?

- A. He should wait for 5 minutes.
- B. Check the product's contact time on the container. ✓ [CORRECT ANSWER]
- C. He should not wait; he should seat his next patient immediately to stay on schedule.
- D. He should go ask Sarah.

Correct Answer: Option B

Clinical Rationale: B. Check the product's contact time on the container. Correct, the product label will list the contact time.

Question 4 [INDIVIDUAL QUESTION]

Which of these groups provides evidence-based recommendations?

- A. The Occupational Safety and Health Administration (OSHA)
- B. The Centers for Disease Control and Prevention (CDC) ✓ [CORRECT ANSWER]**
- C. The Environmental Protection Agency (EPA)
- D. The Food and Drug Administration (FDA)

Correct Answer: Option B

Clinical Rationale: The CDC is an advisory body that provides evidence-based recommendations and guidelines for disease prevention.

Question 5 [CASE-BASED QUESTION]

Case A: Jean, a registered dental hygienist (RDH) working through a temporary agency, is assigned a shift at Bright Smile Dental office. When she arrived, Jean was shown her operatory, schedule, and instruments. She was given a brief tour of the office to show her the radiology room, the sterilization lab, and the break room. No one explained any standard operating procedures (SOPs) or showed her where the OSHA manual was located. As Jean was unloading the autoclave, she noticed the instrument pouches were still wet because another staff member opened the door early because she needed extraction instruments right away.

Wet instruments can increase microbial contamination through a process called _____.

- A. sterilization
- B. contamination
- C. indirect contact
- D. wicking ✓ [CORRECT ANSWER]**

Correct Answer: Option D

Clinical Rationale: D. Wicking is the correct term

Question 6 [CASE-BASED QUESTION]

Case B: Pedro, a registered dental hygienist (RDH), was recently hired at Oceanside Dental Group. On his first day, Pedro was being trained by the lead dental hygienist and Infection Control Coordinator, Sarah. Sarah made sure to explain the schedule, reviewed the OSHA manual, reviewed the walkout procedures, and reviewed all of the infection control policies and procedures located in the Standard Operating Procedures (SOP) manual. As Pedro was turning his treatment room over after dismissing his first patient, he forgot a few things Sarah said regarding the contact time for the disinfectant spray and where the water line treatment tablets were located. He asked another staff member, Cindy, about these questions. Cindy replied, "Sarah goes overboard with infection control. We don't listen to her. Our schedules are so full that we don't have time to do all that stuff." What should Pedro do?

Where could Pedro go to find the location of the water line treatment tablets?

- A. Go ask Sarah.
- B. Go ask the front desk receptionist.
- C. Look in the OSHA manual.
- D. Look in the SOP manual. ✓ [CORRECT ANSWER]**

Correct Answer: Option D

Clinical Rationale: D: Look in the SOP manual.----- Go ask Sarah or go ask the front desk receptionist Incorrect, Pedro needs to learn and abide by the policies and procedures that are outlined. Look in the OSHA manual. Incorrect, the OSHA manual does not contain this information.

Question 7 [INDIVIDUAL QUESTION]

Which of these strains of Hepatitis is NOT bloodborne?

- A. Hepatitis A ✓ [CORRECT ANSWER]
- B. Hepatitis B
- C. Hepatitis C
- D. Hepatitis D

Correct Answer: Option A

Clinical Rationale: Hepatitis A is transmitted via the fecal-oral route. Hepatitis B, C, and D are bloodborne pathogens.

Question 8 [CASE-BASED QUESTION]

Case A: Jean, a registered dental hygienist (RDH) working through a temporary agency, is assigned a shift at Bright Smile Dental office. When she arrived, Jean was shown her operatory, schedule, and instruments. She was given a brief tour of the office to show her the radiology room, the sterilization lab, and the break room. No one explained any standard operating procedures (SOPs) or showed her where the OSHA manual was located. As Jean was unloading the autoclave, she noticed the instrument pouches were still wet because another staff member opened the door early because she needed extraction instruments right away.

How could this practice improve sterilization processes when temporary employees are working?

- A. Have an office manager
- B. Have written Standard Operating Procedures (SOP's) ✓ [CORRECT ANSWER]
- C. Have an OSHA manual
- D. Have OSHA training annually

Correct Answer: Option B

Clinical Rationale: Have written Standard Operating Procedures (SOPs) (correct, written procedures are recommended for training and compliance)

Question 9 [CASE-BASED QUESTION]

Case B: Pedro, a registered dental hygienist (RDH), was recently hired at Oceanside Dental Group. On his first day, Pedro was being trained by the lead dental hygienist and Infection Control Coordinator, Sarah. Sarah made sure to explain the schedule, reviewed the OSHA manual, reviewed the walkout procedures, and reviewed all of the infection control policies and procedures located in the Standard Operating Procedures (SOP) manual. As Pedro was turning his treatment room over after dismissing his first patient, he forgot a few things Sarah said regarding the contact time for the disinfectant spray and where the water line treatment tablets were located. He asked another staff member, Cindy, about these questions. Cindy replied, "Sarah goes overboard with infection control. We don't listen to her. Our schedules are so full that we don't have time to do all that stuff." What should Pedro do?

All of the following are being done correctly with regard to infection control and safety EXCEPT one. Which is the exception?

- A. The office has an infection control coordinator (ICC), Sarah.
- B. The office has written standard operating procedures (SOPs).
- C. The team utilizes a checklist to ensure policies and procedures are being conducted properly. ✓ [CORRECT ANSWER]
- D. The office has training for new employees.

Correct Answer: Option C

Clinical Rationale: C: The team utilizes a checklist to ensure policies and procedures are being conducted properly. This team is NOT using a checklist to ensure policies and procedures are being done properly as evidenced by Cindy's response to Pedro.

Question 10 [CASE-BASED QUESTION]

Case A: Jean, a registered dental hygienist (RDH) working through a temporary agency, is assigned a shift at Bright Smile Dental office. When she arrived, Jean was shown her operatory, schedule, and instruments. She was given a brief tour of the office to show her the radiology room, the sterilization lab, and the break room. No one explained any standard operating procedures (SOPs) or showed her where the OSHA manual was located. As Jean was unloading the autoclave, she noticed the instrument pouches were still wet because another staff member opened the door early because she needed extraction instruments right away.

Using wet instruments could result in _____?

- A. bacterial contamination ✓ [CORRECT ANSWER]
- B. sterility
- C. normal practice
- D. Legionnaire's disease

Correct Answer: Option A

Clinical Rationale: Bacterial contamination is correct and called wicking

Question 11 [INDIVIDUAL QUESTION]

Which of these is considered a critical item according to the Spaulding classification?

- A. Blood pressure cuff
- B. Mouth mirror
- C. Radiograph sensor
- D. Dental hygiene curette ✓ [CORRECT ANSWER]

Correct Answer: Option D

Clinical Rationale: Dental hygiene curettes penetrate soft tissue and bone, classifying them as critical items that require heat sterilization.

Question 12 [CASE-BASED QUESTION]

Case B: Pedro, a registered dental hygienist (RDH), was recently hired at Oceanside Dental Group. On his first day, Pedro was being trained by the lead dental hygienist and Infection Control Coordinator, Sarah. Sarah made sure to explain the schedule, reviewed the OSHA manual, reviewed the walkout procedures, and reviewed all of the infection control policies and procedures located in the Standard Operating Procedures (SOP) manual. As Pedro was turning his treatment room over after dismissing his first patient, he forgot a few things Sarah said regarding the contact time for the disinfectant spray and where the water line treatment tablets were located. He asked another staff member, Cindy, about these questions. Cindy replied, "Sarah goes overboard with infection control. We don't listen to her. Our schedules are so full that we don't have time to do all that stuff." What should Pedro do?

Where could Pedro find the Safety Data Sheet (SDS) for the chemical disinfectant?

- A. The SOP documents
- B. Go ask Sarah
- C. The OSHA manual ✓ [CORRECT ANSWER]
- D. Google

Correct Answer: Option C

Clinical Rationale: The OSHA manual is correct. OSHA requires SDS be located in the OSHA manual

Question 13 [INDIVIDUAL QUESTION]

Which of these is considered a semicritical item according to the Spaulding classification?

- A. Blood pressure cuff
- B. Mouth mirror ✓ [CORRECT ANSWER]**
- C. Extraction forceps
- D. Dental hygiene curette

Correct Answer: Option B

Clinical Rationale: Mouth mirrors touch mucous membranes without penetrating soft tissue or bone, classifying them as semicritical items.

Question 14 [INDIVIDUAL QUESTION]

Which of these is considered a noncritical item according to the Spaulding classification?

- A. Blood pressure cuff ✓ [CORRECT ANSWER]**
- B. Mouth mirror
- C. Radiograph sensor
- D. Dental hygiene curette

Correct Answer: Option A

Clinical Rationale: Blood pressure cuffs touch intact skin only, classifying them as noncritical items requiring low- to intermediate-level disinfection.

Question 15 [INDIVIDUAL QUESTION]

A sneeze or a cough is an example of:

- A. Source of microbe or pathogen
- B. A portal of entry
- C. A portal of exit ✓ [CORRECT ANSWER]**
- D. A vulnerable host

Correct Answer: Option C

Clinical Rationale: A sneeze or cough releases respiratory droplets into the environment, acting as a portal of exit from the source patient.

Question 16 [INDIVIDUAL QUESTION]

All of these are work restrictions for a dental health care worker EXCEPT one. Which is the EXCEPTION?

- A. Conjunctivitis
- B. Pediculosis
- C. Measles
- D. Headache ✓ [CORRECT ANSWER]**

Correct Answer: Option D

Clinical Rationale: A headache is not a CDC work restriction. Active conjunctivitis, pediculosis, measles, and varicella require work restriction.

Question 17 [INDIVIDUAL QUESTION]

Which best describes engineering controls?

- A. Devices to reduce the risk of exposure incidents ✓ [CORRECT ANSWER]
- B. Actions or behaviors that reduce the risk of exposure incidents
- C. Personal protective equipment that reduce the risk of exposure incidents
- D. All of the above

Correct Answer: Option A

Clinical Rationale: Engineering controls are safety devices (such as sharps containers and self-sheathing needles) that isolate or remove bloodborne hazards.

Question 18 [INDIVIDUAL QUESTION]

All of the following are examples of work practice controls EXCEPT one. Which is the EXCEPTION?

- A. Self-sheathing needles ✓ [CORRECT ANSWER]
- B. One-handed scoop technique for recapping a needle
- C. Maintaining a 14–18-inch working distance to reduce droplets
- D. Fulcruming one to four teeth away from tooth being treated

Correct Answer: Option A

Clinical Rationale: Self-sheathing needles are engineering control devices. Work practice controls are behaviors (such as one-handed scoop recapping).

Question 19 [INDIVIDUAL QUESTION]

According to transmission-based precautions, where would a patient with active tuberculosis be treated?

- A. In the doctor's treatment room
- B. In the dental hygienist's operator
- C. In an airborne isolation room ✓ [CORRECT ANSWER]
- D. In a hospital dental clinic

Correct Answer: Option C

Clinical Rationale: Patients with active tuberculosis must be treated in an airborne infection isolation room with negative pressure and N95 respirators.

Question 20 [INDIVIDUAL QUESTION]

According to standard precautions, all of the following are considered potentially infectious EXCEPT one. Which is the EXCEPTION?

- A. Blood
- B. Nonintact skin
- C. Sweat ✓ [CORRECT ANSWER]
- D. Mucous membranes

Correct Answer: Option C

Clinical Rationale: Under Standard Precautions, sweat is NOT considered potentially infectious. Blood, mucous membranes, non-intact skin, and all other body fluids are treated as infectious.

Question 21 [INDIVIDUAL QUESTION]

Which of the following immunizations are recommended for dental health care workers?

- A. Shingles
- B. Hepatitis A and B ✓ [CORRECT ANSWER]
- C. Pneumonia
- D. Polio

Correct Answer: Option B

Clinical Rationale: Hepatitis B immunization is strongly recommended and mandated for dental healthcare workers by OSHA and CDC guidelines.

Question 22 [INDIVIDUAL QUESTION]

When should antibody follow-up testing occur for the Hepatitis B vaccine?

- A. 1–2 months after the final injection ✓ [CORRECT ANSWER]
- B. 3–4 months after the final injection
- C. 4–6 months after the final injection
- D. 1–2 years after the final injection

Correct Answer: Option A

Clinical Rationale: Post-vaccination serologic antibody testing (anti-HBs titer) should be performed 1 to 2 months after the final (3rd) Hepatitis B injection.

Question 23 [INDIVIDUAL QUESTION]

Hand hygiene should be performed at all of these times EXCEPT one. Which is the EXCEPTION?

- A. Before gloving (donning)
- B. After gloves are removed (doffing)
- C. After using the restroom
- D. Before using the restroom ✓ [CORRECT ANSWER]

Correct Answer: Option D

Clinical Rationale: Hand hygiene is required before gloving, after removing gloves, and after using the restroom, but NOT before using the restroom.

Question 24 [INDIVIDUAL QUESTION]

Choose the correct sequence for donning personal protective equipment (PPE).

- A. Protective clothing (lab coat or gown), mask & eyewear, hand hygiene, then gloves ✓ [CORRECT ANSWER]
- B. Mask & eyewear, hand hygiene and gloves, then protective clothing (lab coat or gown)
- C. Protective clothing (lab or gown), hand hygiene and gloves, then mask & eyewear
- D. Mask & eyewear, protective clothing (lab coat or gown), then hand hygiene and gloves

Correct Answer: Option A

Clinical Rationale: The correct donning sequence for PPE is: protective clothing (gown), mask & eyewear, hand hygiene, and lastly gloves.

Question 25 [INDIVIDUAL QUESTION]

Which types of gloves should be used during instrument reprocessing?

- A. Nonsterile examination gloves
- B. Sterile surgical gloves
- C. Utility gloves ✓ [CORRECT ANSWER]
- D. Overgloves (food handler's gloves)

Correct Answer: Option C

Clinical Rationale: Heavy-duty utility gloves are required during instrument cleaning and reprocessing because they are puncture- and chemical-resistant.

Question 26 [INDIVIDUAL QUESTION]

Latex allergy is which type of reaction?

- A. Type I immediate ✓ [CORRECT ANSWER]
- B. Type II Cytotoxic
- C. Type III Immune Complex
- D. Type IV delayed

Correct Answer: Option A

Clinical Rationale: Natural rubber latex protein allergy is a Type I immediate IgE-mediated hypersensitivity reaction.

Question 27 [INDIVIDUAL QUESTION]

Which type of surface disinfectant should be used to wipe clinical contact surfaces that might be contaminated with blood or other potentially infectious material (OPIM)?

- A. High level
- B. Intermediate level ✓ [CORRECT ANSWER]
- C. Low level
- D. None of the above

Correct Answer: Option B

Clinical Rationale: Clinical contact surfaces contaminated with blood or OPIM require an intermediate-level disinfectant with a tuberculocidal claim.

Question 28 [INDIVIDUAL QUESTION]

Intermediate-level disinfectants carry all of these kill claims EXCEPT one. Which is the EXCEPTION?

- A. C. difficile ✓ [CORRECT ANSWER]
- B. HIV
- C. Hepatitis B
- D. S. aureus

Correct Answer: Option A

Clinical Rationale: Intermediate-level disinfectants do NOT kill bacterial endospores such as *Clostridioides difficile*.

Question 29 [INDIVIDUAL QUESTION]

Which of the following is NOT considered a clinical contact surface?

- A. Chair controls
- B. Bracket tray
- C. Floors ✓ [CORRECT ANSWER]
- D. Saliva ejector control

Correct Answer: Option C

Clinical Rationale: Floors are housekeeping surfaces, not clinical contact surfaces.

Question 30 [INDIVIDUAL QUESTION]

Which organization regulates chemical disinfectants?

- A. FDA
- B. CDC
- C. OSHA
- D. EPA ✓ [CORRECT ANSWER]

Correct Answer: Option D

Clinical Rationale: The Environmental Protection Agency (EPA) regulates chemical disinfectants and environmental sterilants in the United States.

Question 31 [INDIVIDUAL QUESTION]

Dental Unit Water Lines measured in colony-forming units per milliliter (CFU/mL) must not exceed which level of bacterial contamination?

- A. 50 CFU/mL
- B. 100 CFU/mL
- C. 500 CFU/mL ✓ [CORRECT ANSWER]
- D. 1000 CFU/mL

Correct Answer: Option C

Clinical Rationale: Dental unit water must not exceed ≤ 500 CFU/mL of heterotrophic water bacteria, meeting EPA drinking water standards.

Question 32 [INDIVIDUAL QUESTION]

What is the first action that should be taken when a percutaneous (sharps injury) happens?

- A. Wash the area with soap and water ✓ [CORRECT ANSWER]
- B. Yell for help
- C. Report the injury
- D. Get an HIV test

Correct Answer: Option A

Clinical Rationale: The immediate first step following a percutaneous sharps injury is to wash the affected area thoroughly with soap and water.

Question 33 [INDIVIDUAL QUESTION]

Where should blood-soaked or blood-saturated items be discarded?

- A. In the trash can at the front desk
- B. In the operator
- C. In a biohazard container ✓ [CORRECT ANSWER]
- D. In the sterilization area trash can

Correct Answer: Option C

Clinical Rationale: Blood-soaked or blood-saturated items (capable of releasing liquid blood when compressed) must be discarded in a biohazard container.

Question 34 [INDIVIDUAL QUESTION]

How soon should a dental health care worker seek medical care after an exposure incident?

- A. Within one hour
- B. Within two hours ✓ [CORRECT ANSWER]
- C. Within four hours
- D. Within 24 hours

Correct Answer: Option B

Clinical Rationale: Medical evaluation and Post-Exposure Prophylaxis (PEP) for HIV should be initiated as soon as possible, ideally within 2 hours of exposure.

Question 35 [INDIVIDUAL QUESTION]

All of the following are acceptable methods of cleaning for removing debris or bioburden from instruments prior to sterilization EXCEPT one. Which is the EXCEPTION?

- A. Hand scrubbing with a short-handled brush ✓ [CORRECT ANSWER]
- B. An ultrasonic bath
- C. An instrument washer
- D. An instrument washer/disinfector

Correct Answer: Option A

Clinical Rationale: Hand scrubbing with a short-handled brush is NOT recommended due to high risk of sharps injuries. Automated cleaning or long-handled brushes should be used.

Question 36 [INDIVIDUAL QUESTION]

Autoclave tape is an example of which type of chemical indicator?

- A. Class I ✓ [CORRECT ANSWER]
- B. Class II
- C. Class III
- D. Class VI

Correct Answer: Option A

Clinical Rationale: Autoclave tape is a Class I process indicator (external single-parameter indicator verifying exposure to heat).

Question 37 [INDIVIDUAL QUESTION]

The best place to store instruments after sterilization is _____?

- A. unwrapped in the sterilization drawer.
- B. unwrapped in the dental operator.
- C. wrapped in a drawer in the operator. ✓ [CORRECT ANSWER]
- D. wrapped on a shelf above the autoclave.

Correct Answer: Option C

Clinical Rationale: Instruments must remain wrapped in sterile packaging and stored in clean, dry drawers or cabinets until point of use.

Question 38 [INDIVIDUAL QUESTION]

How frequently should biological monitoring (spore testing) occur?

- A. At least every hour
- B. At least daily
- C. At least weekly ✓ [CORRECT ANSWER]
- D. At least monthly

Correct Answer: Option C

Clinical Rationale: Biological monitoring (spore testing) of sterilizers must be performed at least weekly per CDC recommendations.

Question 39 [INDIVIDUAL QUESTION]

Which type of chemical indicator should be placed inside of a wrapped instrument cassette?

- A. Class I
- B. Class II
- C. Class IV
- D. Class V ✓ [CORRECT ANSWER]

Correct Answer: Option D

Clinical Rationale: Class V chemical integrators react to all critical sterilization parameters (time, temperature, steam) and should be placed inside instrument cassettes.

Question 40 [INDIVIDUAL QUESTION]

All of the following methods inhibit the formation of biofilm in dental unit water lines EXCEPT one. Which is the EXCEPTION?

- A. Flushing water lines for 2 minutes ✓ [CORRECT ANSWER]
- B. Independent water reservoir
- C. Chemical disinfection of water lines
- D. Sterile-water delivery systems

Correct Answer: Option A

Clinical Rationale: Flushing water lines for 2 minutes removes loose planktonic bacteria but does NOT remove or inhibit waterline biofilm formation.

Question 41 [INDIVIDUAL QUESTION]

Which governmental agency is responsible for approving the safety and effectiveness of small office sterilizers and ultrasonic units?

- A. EPA
- B. FDA ✓ [CORRECT ANSWER]
- C. CDC
- D. OSHA

Correct Answer: Option B

Clinical Rationale: The Food and Drug Administration (FDA) regulates medical devices, including office sterilizers, ultrasonic units, and gloves.

Question 42 [INDIVIDUAL QUESTION]

What is the ultimate goal of disease prevention in the oral health care (OHC) setting?

- A. Prevent pathogens from entering the OHC setting.
- B. Sterilize clinical contact surfaces between patients.
- C. Prevent OHC related infections and disease. ✓ [CORRECT ANSWER]
- D. Eliminate all microbes in the OHC setting.

Correct Answer: Option C

Clinical Rationale: The ultimate goal of infection control in oral healthcare is to prevent healthcare-associated infections and disease transmission.

Question 43 [INDIVIDUAL QUESTION]

A dental health care worker has occupationally acquired tuberculosis. Which of the following modes of transmission would be implicated in this case?

- A. Direct contact
- B. Indirect contact
- C. Droplet infection
- D. Airborne infection ✓ [CORRECT ANSWER]

Correct Answer: Option D

Clinical Rationale: Mycobacterium tuberculosis is transmitted via airborne infection when infectious microscopic droplets are inhaled into the lungs.

Question 44 [INDIVIDUAL QUESTION]

Which term is used for the practice of treating a patient's blood, body fluids, nonintact skin, and mucous membranes as potentially infectious?

- A. Standard precautions ✓ [CORRECT ANSWER]
- B. Pervasive precautions
- C. Universal precautions
- D. Protective precautions

Correct Answer: Option A

Clinical Rationale: Standard Precautions expands Universal Precautions to treat blood, all body fluids (except sweat), nonintact skin, and mucous membranes as infectious.

Question 45 [INDIVIDUAL QUESTION]

Today, effective vaccines are available to prevent which of the following infections?

- A. HIV
- B. Hepatitis C
- C. Hepatitis B ✓ [CORRECT ANSWER]
- D. Herpes simplex

Correct Answer: Option C

Clinical Rationale: Effective preventive vaccines exist for Hepatitis B (and Hepatitis A). No vaccines currently exist for HIV, HCV, or Herpes simplex.

Question 46 [INDIVIDUAL QUESTION]

Which of the following is an example of an engineering control?

- A. Eyewear
- B. Sharps container ✓ [CORRECT ANSWER]
- C. Using a one-handed scoop technique for recapping a needle
- D. Handwashing

Correct Answer: Option B

Clinical Rationale: A sharps container is a safety device and an example of an engineering control.

Question 47 [INDIVIDUAL QUESTION]

Which of the following is not a link in the chain of infection?

- A. A pathogen in sufficient number
- B. A mode of transmission for the pathogen
- C. An immune host ✓ [CORRECT ANSWER]
- D. A portal of entry into a new host
- E. A reservoir for the pathogen

Correct Answer: Option C

Clinical Rationale: An immune host breaks the chain of infection. The chain of infection link is a SUSCEPTIBLE host.

Question 48 [INDIVIDUAL QUESTION]

A person who is immune/recovered from a disease but can still transmit the disease is called:

- A. A susceptible host
- B. A source patient
- C. A carrier ✓ [CORRECT ANSWER]
- D. An at-risk dental health care worker

Correct Answer: Option C

Clinical Rationale: A carrier is an individual who has recovered or is asymptomatic but continues to harbor and transmit infectious pathogens.

Question 49 [INDIVIDUAL QUESTION]

All of the following tasks should be conducted to maintain high-quality dental unit water EXCEPT one. Which is the EXCEPTION?

- A. Use heated water for patient comfort. ✓ [CORRECT ANSWER]**
- B. Flush water lines for 20–30 seconds between patients.
- C. Use low-level chemical germicides in dental unit water lines.
- D. Periodically test the CFU/mL level in dental unit water.

Correct Answer: Option A

Clinical Rationale: Heated water is NOT recommended in dental unit waterlines because warmth promotes rapid microbial biofilm proliferation.

Question 50 [INDIVIDUAL QUESTION]

The Centers for Disease Control and Prevention recommends that all dental offices have a designated person who handles infection control training records, monitors infection control policies and procedures, and who communicates updates about infection control policies and procedures. This person is known as the _____.

- A. OSHA monitor
- B. Infection control coordinator ✓ [CORRECT ANSWER]**
- C. Infection Manager
- D. CDC Monitor

Correct Answer: Option B

Clinical Rationale: The CDC recommends designating an Infection Control Coordinator (ICC) to manage training, policies, and compliance in every dental practice.